**GROUP 4.2**

**Eetu Knutars**

**Eero Ryhänen**

**Lauri Heininen**

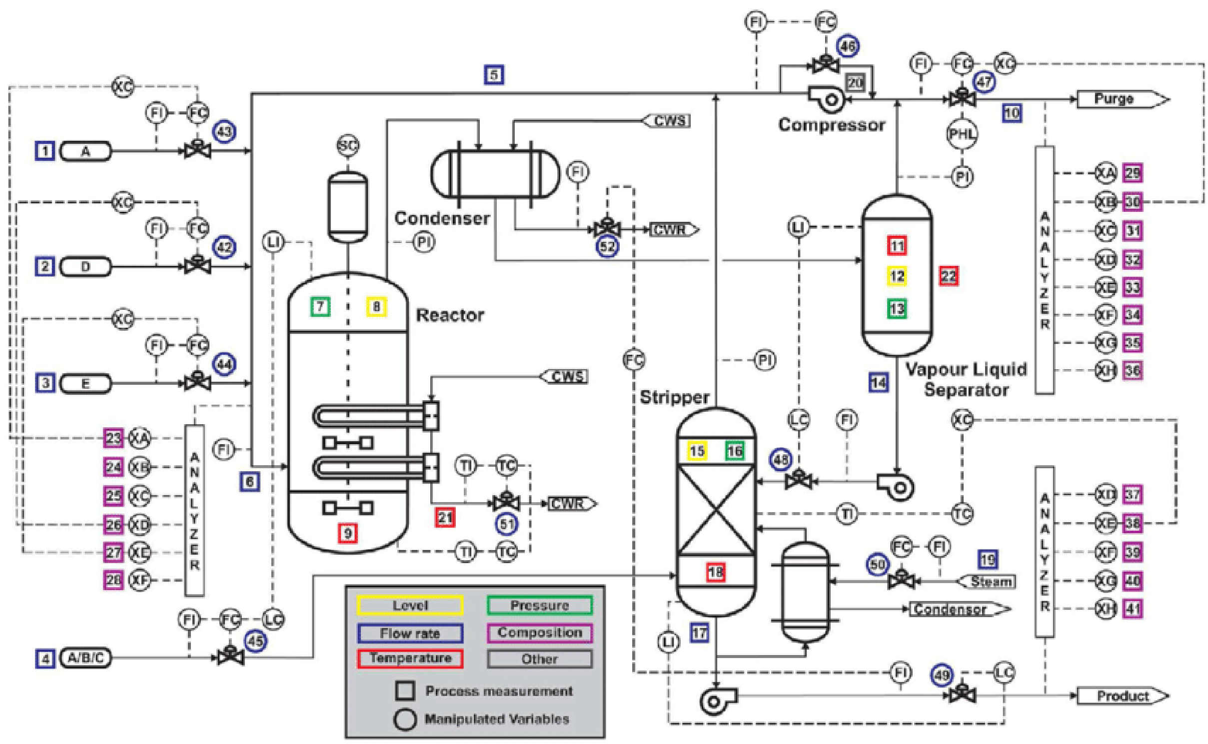
**Eetu Peltola**

**ADAML Workshop 18.9.2025**

**Note! For every task, present <u>at least</u> one plot**, that will be commented. The code doesn't give you any points. It is provided.

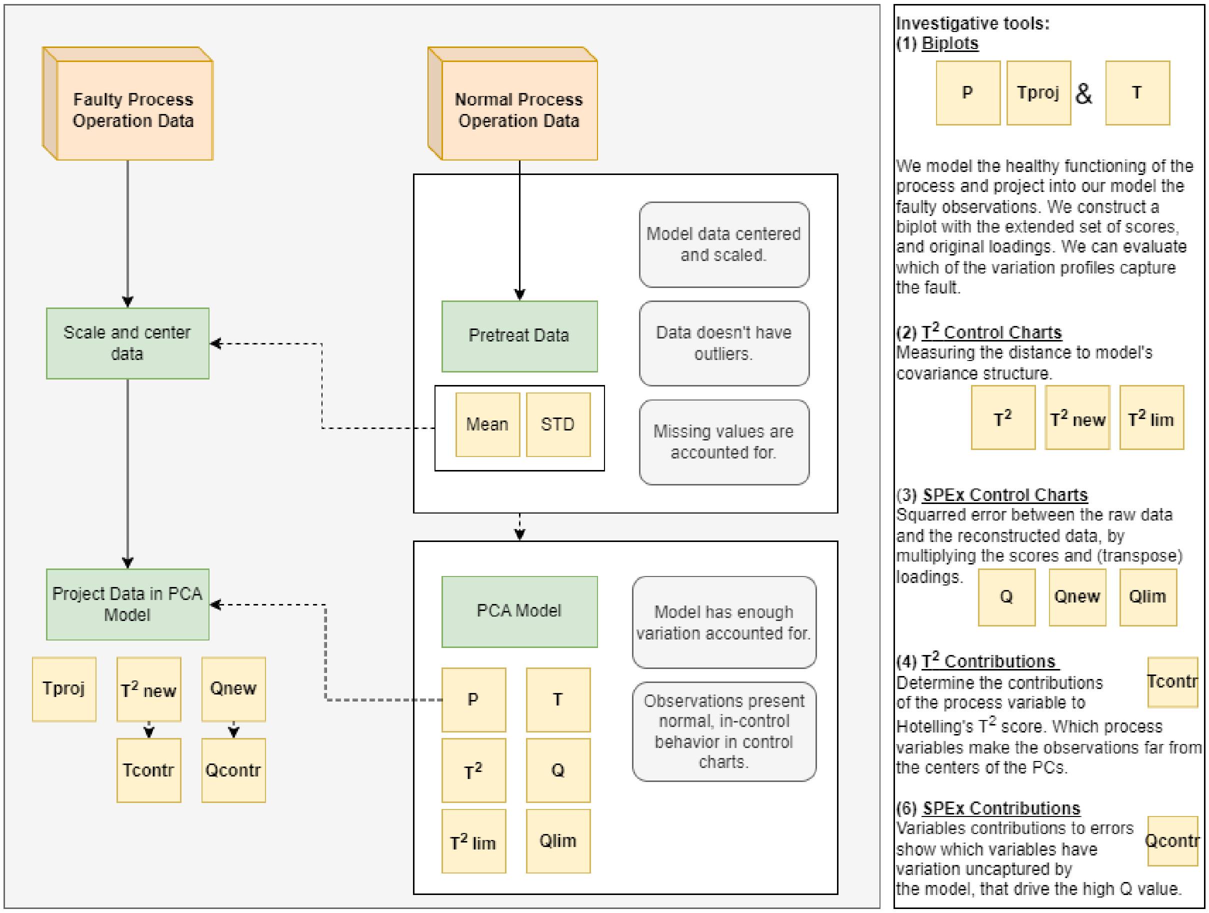
**Dataset Description**

The TEP is comprised of 8 chemical components in total: 4 reactants, 2 products, 1 by-product and 1 inert component. These components undergo a chemical process dominated by 5 main process units: a reactor that allows for the reaction of the gaseous feed components (A, C, D and E) into liquid products (G and H), a condenser to cool-down the gaseous product stream coming out of the reactor, a gas-liquid separator to split gas and liquid components from the cooled product stream, a centrifugal compressor to flow this gas stream back into the reactor and a stripper to handle the efficient separation of the 2 products from any unreacted feed components.



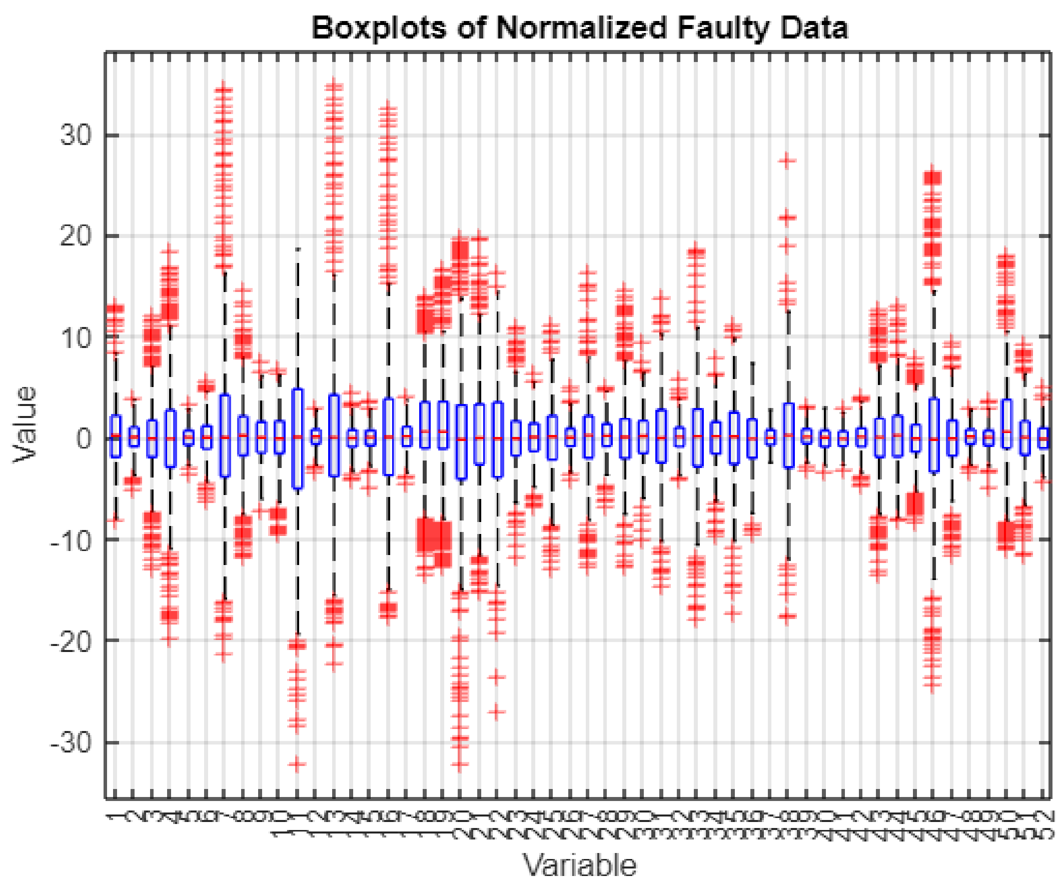

The variable description is seen in the table below:

| Process measurement | | Composition measurement | | Manipulated variable | |
|---|---|---|---|---|---|
| Variable | Description | Variable | Description | Variable | Description |
| 1 | A Feed | 23 | Reactor feed component A | 42 | D feed flow |
| 2 | D Feed | 24 | Reactor feed component B | 43 | E feed flow |
| 3 | E Feed | 25 | Reactor feed component C | 44 | A feed flow |
| 4 | Total Feed | 26 | Reactor feed component D | 45 | Total feed flow |
| 5 | Recycle flow | 27 | Reactor feed component E | 46 | Compressor recycle valve |
| 6 | Reactor feed rate | 28 | Reactor feed component F | 47 | Purge valve |
| 7 | Reactor pressure | 29 | Purge component A | 48 | Separator product liquid flow |
| 8 | Reactor level | 30 | Purge component B | 49 | Stripper product liquid flow |
| 9 | Reactor temperature | 31 | Purge component C | 50 | Stripper steam valve |
| 10 | Purge rate | 32 | Purge component D | 51 | Reactor cooling water flow |
| 11 | Separator temperature | 33 | Purge component E | 52 | Condenser cooling water flow |
| 12 | Separator level | 34 | Purge component F | | |
| 13 | Separator pressure | 35 | Purge component G | | |
| 14 | Separator underflow | 36 | Purge component H | | |
| 15 | Stripper level | 37 | Product component D | | |
| 16 | Stripper pressure | 38 | Product component E | | |
| 17 | Stripper underflow | 39 | Product component F | | |
| 18 | Stripper temperature | 40 | Product component G | | |
| 19 | Stripper steam flow | 41 | Product component H | | |
| 20 | Compressor work | | | | |
| 21 | Reactor cooling water outlet temperature | | | | |
| 22 | Separator cooling water outlet temperature | | | | |




**Task 0.(0p, sorry)** Load `d00_te.dat` (healthy = normal operation) and `d01_te.dat` (faulty) into MATLAB and take a moment to note how many variables and observations you have in each. Standardize both datasets using means and standard deviations computed from the healthy data only, and keep those μ and σ values for later use.

| |
|---|
| Data files were loaded.<br><br>Both data files contained 960 data rows and 52 variables<br><br>Data matrices were normalized to unit variance and zero mean using the mean and variance of the healthy data. |

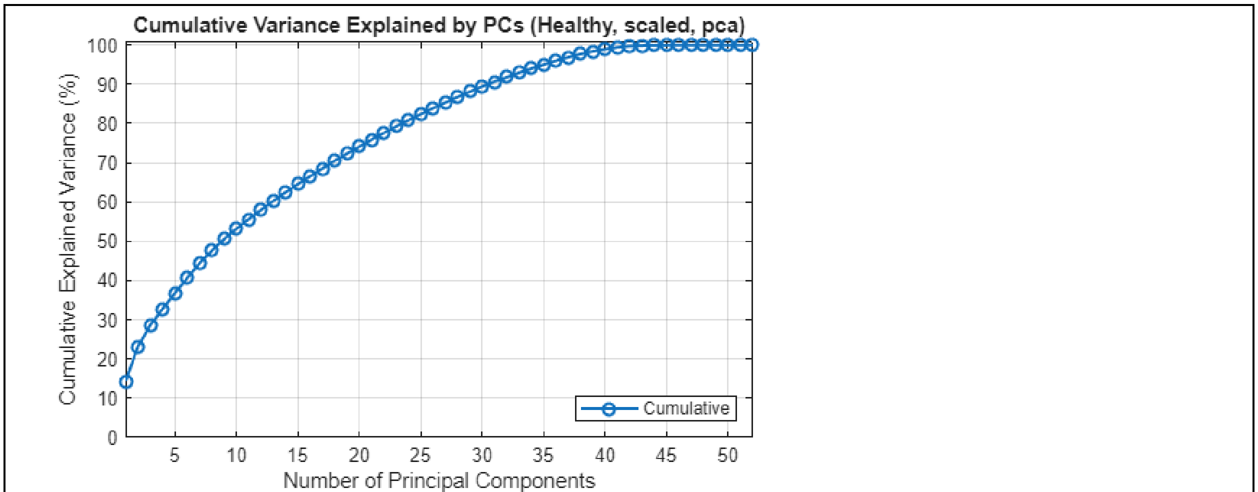


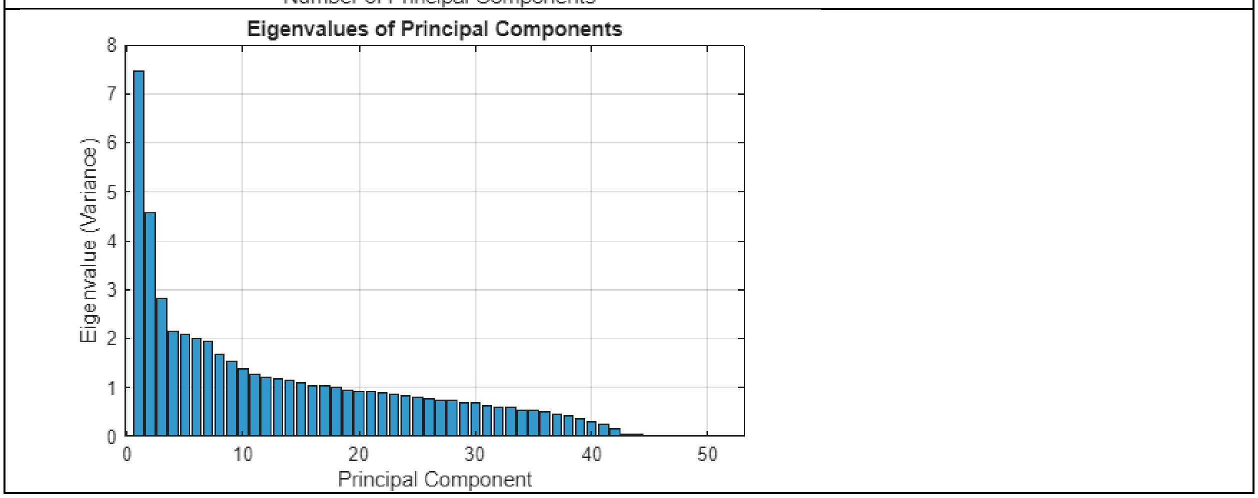

In the figure are the normalized faulty data shown in with a boxplot. The high variance variables like 7, 13 and 16 etc. could be crucial variables to detect the faults.

**Task 1.(1p)** Fit a PCA model to the standardized healthy data. Examine the cumulative explained variance across components and also the eigenvalues (variance per PC) in descending order. Choose the number of principal components, k, based on these plots and process considerations rather than a fixed variance threshold. Since data is auto-scaled, you can use “higher than 1” rule of thumb. You will adjust these values later on if fault is not captured with these parameters.

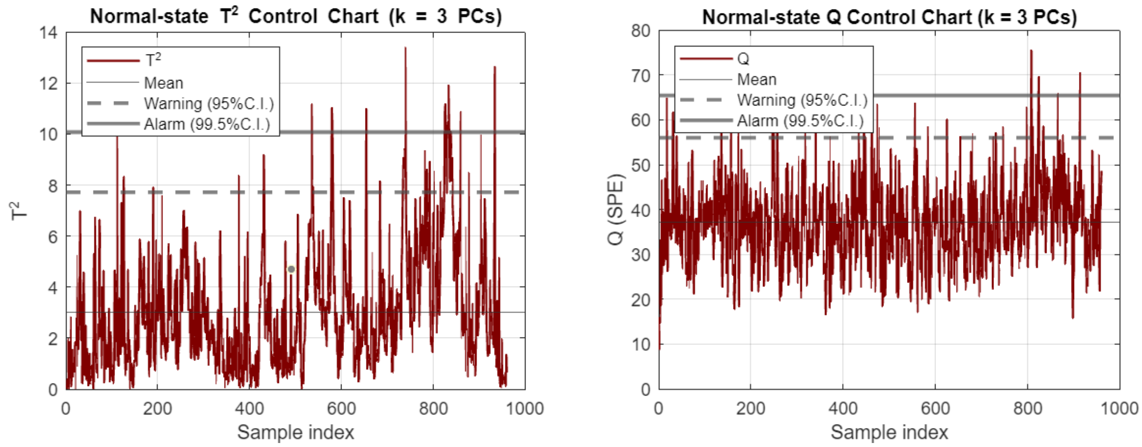
PCA model was fitted to the healthy data. The cumulative variance explained by PCs as well as the eigenvalues (variance) of each PC are shown in plots below.

Based on the variance of each PC we decided to choose k = 3 as the number of principle components used.



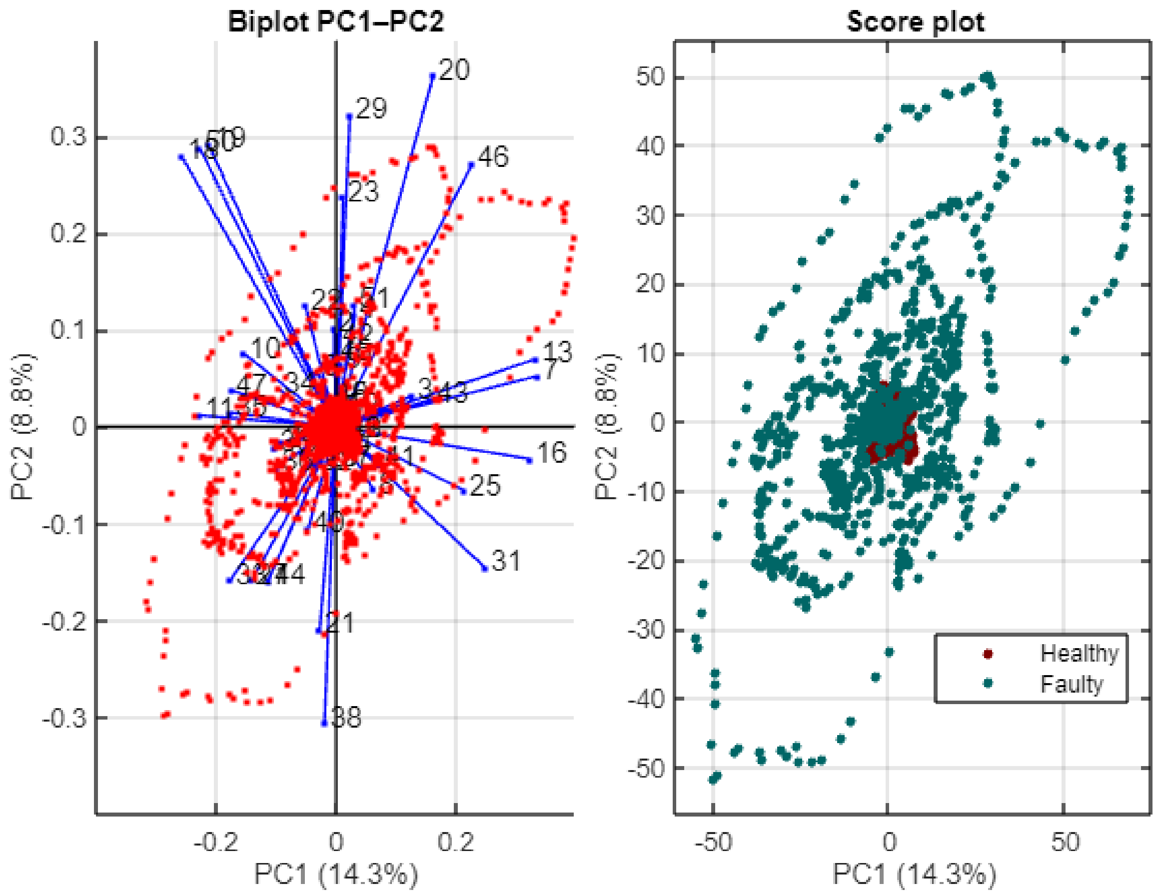




**Task 2. (1p)** Using your chosen number of PCs, build normal-state control charts from the healthy data alone. Compute Hotelling's $T^2$ values and SPEx (also called Q or SPE) for the healthy samples with the provided functions, then estimate empirical limits from the healthy set: the mean as the centerline, the warning limit set to 95%C.I (mean + two standard deviations), and the alarm limit set to 99.5% (mean + three standard deviations). Plot the $T^2$ chart and the SPEx chart for the healthy run with these limits and comment on whether any points appear as outliers (if they do, treat them as potential false alarms here).




At around index 800 we can see a small bump in the Q Control chart and for a short amount of time the values go above alarm level. This can be treated as a false alarm, and the values go back to normal relatively fast after this bump.

**Task 3. (1p)** Project the faulty dataset onto the healthy PCA model. For PC pairs up to PC7 (i.e., 1–2, 2–3, ..., 6–7), analyze if the fault can be captured by the biplots. Select one biplot to present. Comment on (a) clustering of variables (b) what can you tell about the values of these variables in the faulty dataset (c) which variables are positively correlated, which are uncorrelated, and which have a negative correlation. Generally comment on how the faulty samples differ from the normal operating regime and which variables (via loadings and directions) appear to drive the separation.



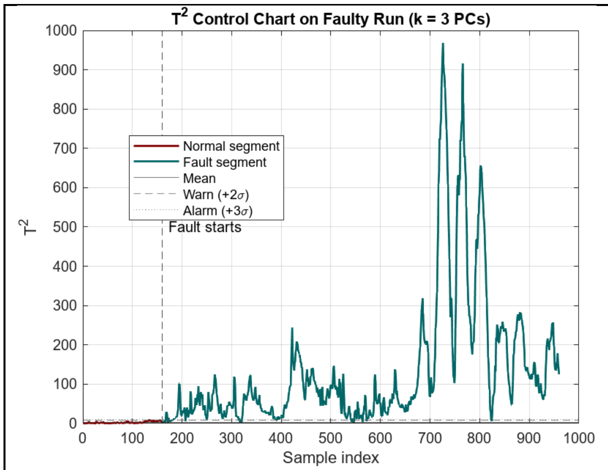

a) Variable clustering can be seen as the distribution of the vectors in the biplot. When comparing PC1 and PC2, the variables are spread quite well around the first two principal components. Some clusters can be noticed where 3+ variables contribute similarly to the principal components, such as the three variables located at around (-0.2, 0.3) in the biplot.

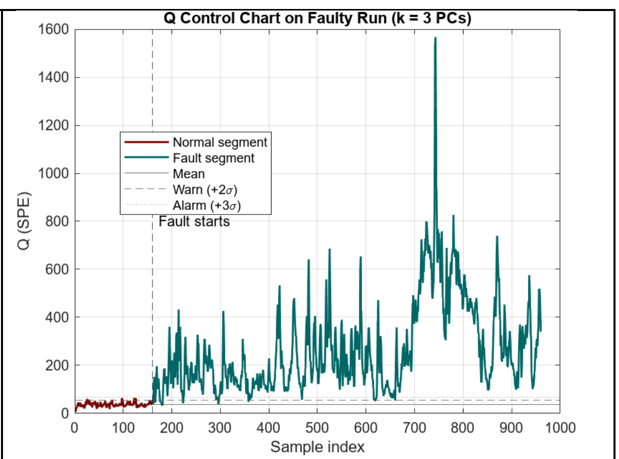
b) In the faulty score plot we can see that in faulty state the scores start differing significantly both in PC1 and PC2. This can tell us that two or more variables, that do not normally correlate with each other under normal operating conditions, are affecting the process simultanenously (correlate in faulty conditions).

c) The variables clustered together in the biplot correlate positively with each other, since they affect similarly to the principal components. Negative correlation can be found with variables that are directed to the opposite directions, for instance in the biplot we can see variables with very little effect to PC1, but which affect to PC2 opposite ways. The variables that are perpendicular to each other in the biplot do not correlate at all with each other

In the score plot we can clearly see how the faulty data differs from the healthy state. Faulty scores circle the centrum widely spread, while the healthy data scores are in the centrum as a nice cluster. In the score plot we can also see a period of time when the faulty score circle widely around the center continuously and systematically, not just randomly. This is likely caused by the faulty state and it's effects to multiple variables.

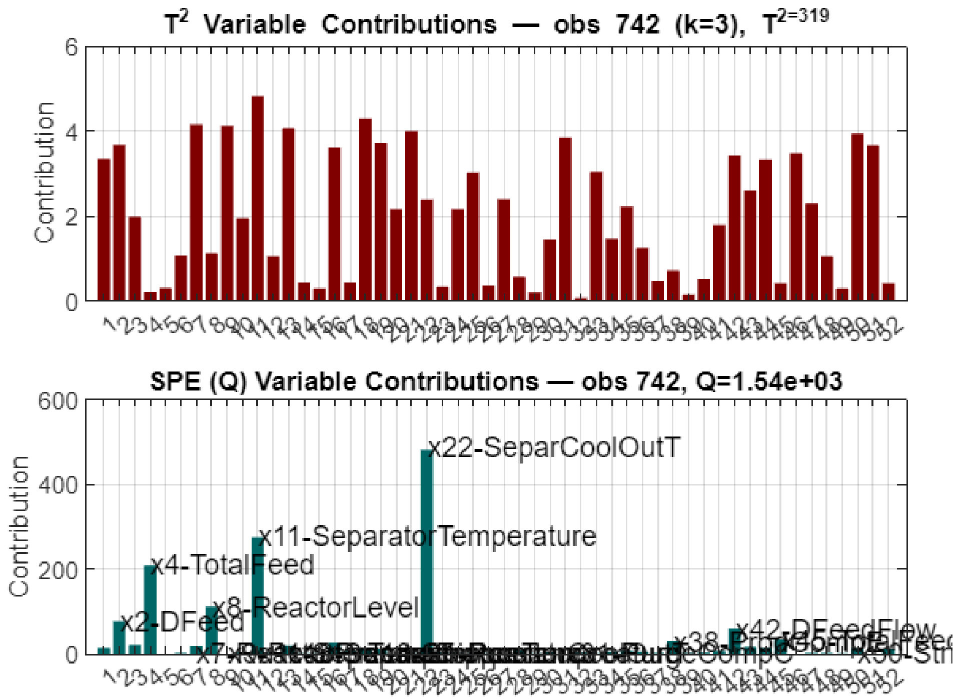
**Task 4**. **(1p)** Add the faulty run to your control charts. Compute $T^2$ and SPEx for the entire faulty run using the same PCA model and overlay them on two separate line charts. Reuse the healthy-state limits from Task 2 and mark the fault start. Identify which chart signals the fault earlier and approximately at which sample index. (Note: if you cannot capture the fault in the no. PCs you have initially chosen, evaluate the control charts using a different number of PCs! You can change your PC choice to the minimal number of PCs that can capture the fault.)



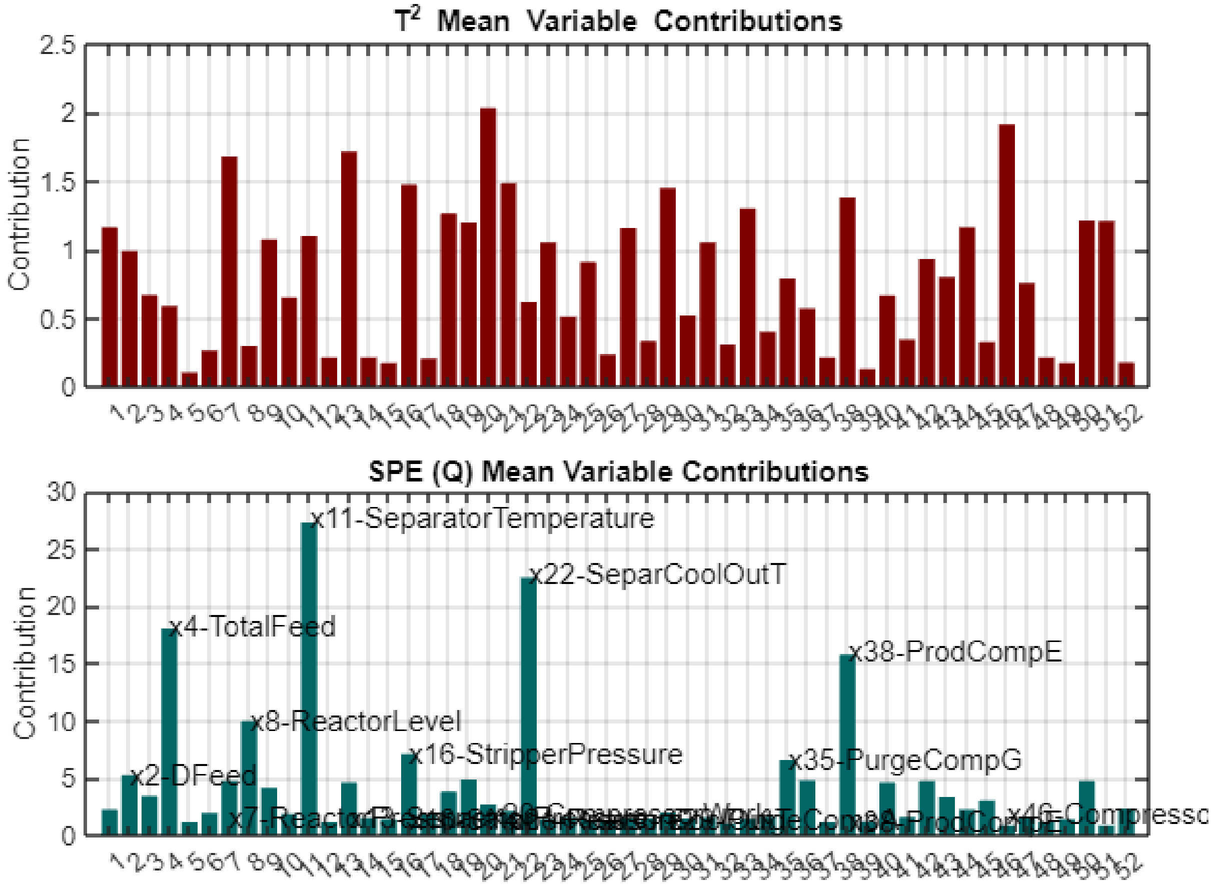

Fault can be identified faster from the Q control chart. Fault is identifiable almost immediately after it happens, around sample index 160. From T2 control chart it can be reliably identified closer to sample index 170.

**Task 5. (1p)** Pick one faulty observation at or after index 161 and diagnose it with variable contributions. Briefly explain which variables contribute most and relate this back to the biplots and loadings from Task 3.



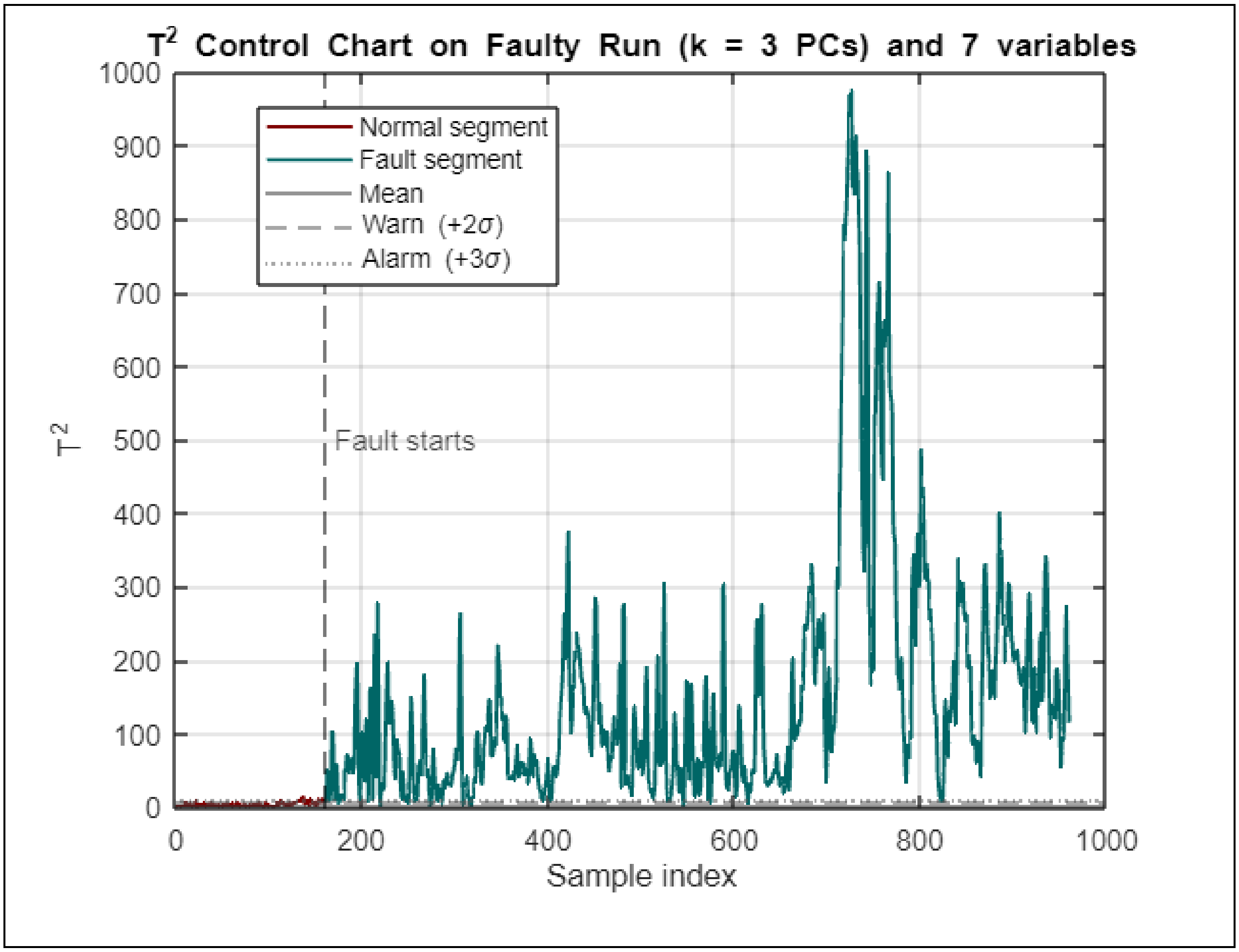

Most significant variables are x22-SeparCoolOutT, x11-SeparatorTemperature, x4-TotaFeed, x8-ReactorLevel and x2-DFeed. The sample was taken at index 742, which had the highest value on the Q control chart.

**Task 6. (1p)** You have identified variables with most contribution to control charts. Delete the most unimportant variables until you reach the minimum number of variables able to capture the fault. Which are the variables most essential in capturing the fault?

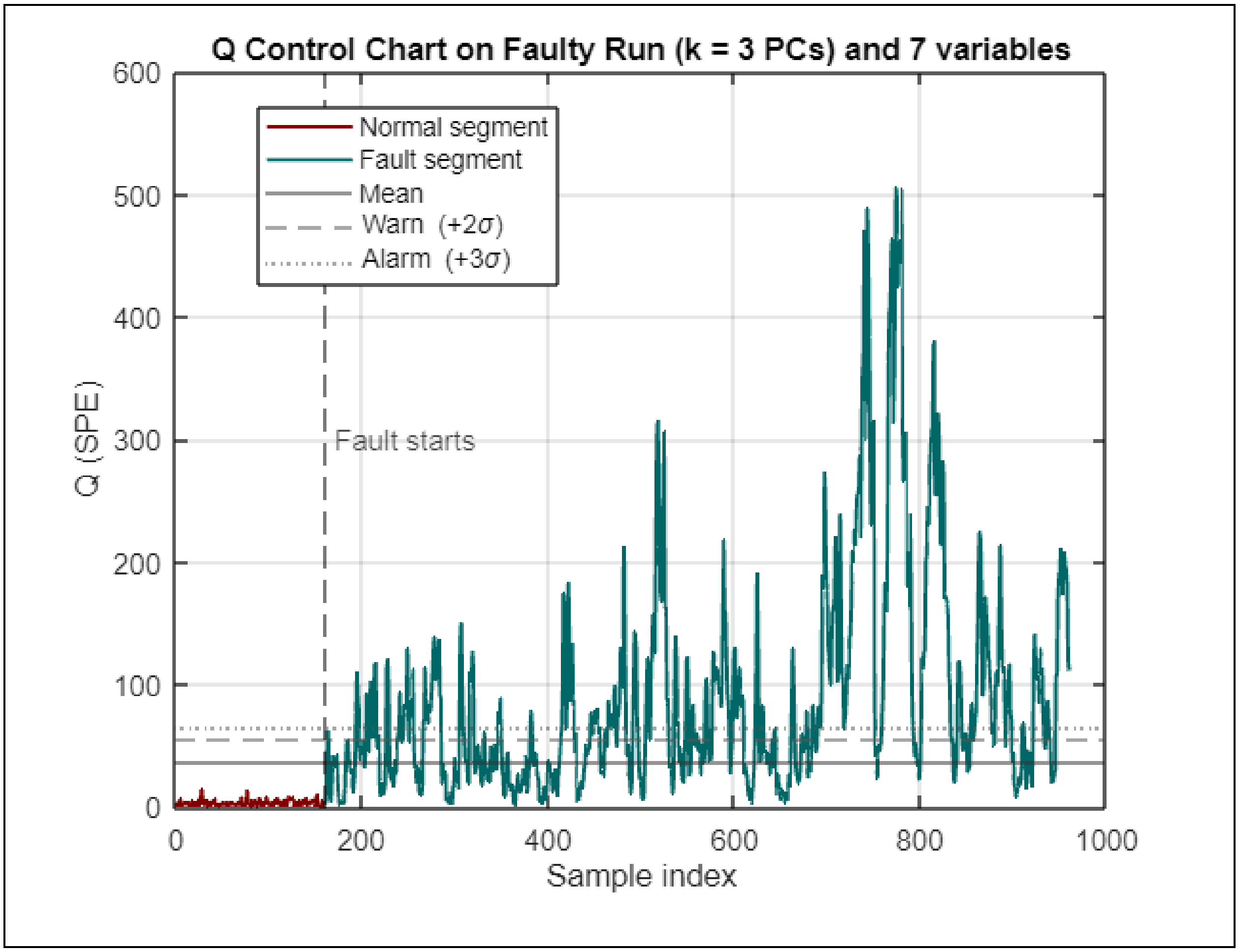



We chose the most important variables by calculating the mean contribution of each variable. With 7 variables we got both of the control charts to give an alarm right after the fault started. With 6 variables the Q chart didn't give an alarm after the fault.

The variables chosen: 2, 4, 8, 11, 22, 35, 38



T2 Control Chart on Faulty Run (k = 3 PCs) and 7 variables
Normal segment
Fault segment
Mean
Warn (+2σ)
Alarm (+3σ)
Fault starts
T2
Sample index
0
200
400
600
800
1000
100
300
500
700
900




**Task 7. (1p) Collaboration with other groups with the same fault number**. Write down the groups you have collaborated, and briefly described (a) what you collaborated on and (c) if they got the same conclusions and results as you.

| |
|---|
| We collaborated with Group 4.1 for understanding the "higher than 1" rule of thumb for selecting the number of principal component. We also collaborated with the group for discussing about the variable clustering in the biplots. We discussed about the issues together and arrived to the same conclusions. |

**Task 8. (1p)** Elect a member of the group (the president) to present one plot and draw conclusions from it. Talk with the other groups and synchronize so that not everyone from the same fault presents the same plot. When time comes, the president is asked to present in front of the classroom.

| |
|---|
| We elected Eero Ryhänen as the group predident to present our plot. |